Blueprint for the Ternary Quartic Rank-4 Proof

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1 Statement and setup

We work in dehomogenized coordinates: a ternary quartic is viewed as a quartic polynomial in two affine variables x_0, x_1. The optimization variable is a rank-4 Burer–Monteiro factor

$$u = (u_0, u_1, u_2, u_3) \in \mathbb{R}[x_0, x_1]^4_{\le 2}.$$

All tangent directions are subject to the same quadratic degree bound.

Definition 1.1 (Basic maps; Lean: `sigma`, `A`, `residual`, `IsSOCP`). For $u, v \in \mathbb{R}[x_0, x_1]^4_{\le 2}$ and $p \in \mathbb{R}[x_0, x_1]_{\le 4}$, define

$$\sigma(u) = \sum_{i=0}^{3} u_i^2, \qquad A_u(v) = \sum_{i=0}^{3} u_i v_i, \qquad r(u,p) = \sigma(u) - p.$$

Fix a positive-definite bilinear form B on $\mathbb{R}[x_0, x_1]$. We say that u is a first-order critical point of p if

$$B\big(A_u(v),\, r(u,p)\big) = 0 \qquad \text{for all } v \in \mathbb{R}[x_0, x_1]^4_{\le 2},$$

and a second-order critical point if, in addition,

$$B\big(\sigma(v),\, r(u,p)\big) + 2B\big(A_u(v), A_u(v)\big) \ge 0 \qquad \text{for all } v \in \mathbb{R}[x_0, x_1]^4_{\le 2}.$$

Definition 1.2 (SOS quartic target; Lean: `IsSOSQuartic`). An *SOS quartic* is a polynomial $p \in \mathbb{R}[x_0, x_1]_{\le 4}$ admitting a finite representation

$$p = \sum_{j=1}^{k} q_j^2, \qquad q_j \in \mathbb{R}[x_0, x_1]_{\le 2}.$$

Definition 1.3 (Relation map and exact affine relations; Lean: `relationPoly`, `relationPolyLin`, `exactAffineSubmodule`). For fixed $u \in \mathbb{R}[x_0, x_1]^4_{\le 2}$, define the relation map

$$\rho_u : \mathbb{R}^4 \to \mathbb{R}[x_0, x_1], \qquad \rho_u(c) = \sum_{i=0}^{3} c_i u_i.$$

This is the linear map called `relationPolyLin` in Lean. Its kernel is the space of scalar linear dependences among the four coordinates of u.

Define the *exact affine relation space*

$$\mathcal{E}(u) = \{c \in \mathbb{R}^4 : \rho_u(c) \in \mathbb{R}[x_0, x_1]_{\le 1}\}.$$

Equivalently, $\mathcal{E}(u)$ is the kernel of the map taking $\rho_u(c)$ to its homogeneous quadratic coefficients.

Remark 1.4. The theorem proved in Lean is the statement that every admissible rank-4 SOCP over an SOS quartic has zero residual:

$$r(u,p) = 0.$$

Equivalently, there are no spurious second-order critical points in the rank-4 ternary quartic problem.

2 The certificate mechanism

The proof follows the certificate philosophy already used in the univariate writeup: one shows that every SOS square q^2 can be written as an image term for A_u plus kernel-square corrections. The SOCP inequalities then force the residual to vanish.

Lemma 2.1 (Abstract image-plus-kernel-square certificate; Lean: `admissible_image_plus_cone_residual_eq_zero`). *Let $u \in \mathbb{R}[x_0, x_1]^4_{\le 2}$ be a* SOCP *for an SOS quartic p. Assume that*

$$p = q_{\mathrm{img}} + \sum_{m=1}^{N} \sigma(w^{(m)}),$$

where $q_{\mathrm{img}} \in \mathrm{im}(A_{u_{\mathrm{img}}})$ for some admissible u_{img}, each $w^{(m)} \in \mathbb{R}[x_0, x_1]^4_{\le 2}$ satisfies $A_u(w^{(m)}) = 0$, and $r(u, p)$ is orthogonal to $\mathrm{im}(A_{u_{\mathrm{img}}})$. Then $r(u, p) = 0$.

Proof. Set $r = r(u, p)$. By first-order criticality, the image term pairs to zero:

$$B(q_{\mathrm{img}}, r) = 0.$$

For each kernel direction $w = w^{(m)}$, the Hessian inequality gives

$$B(\sigma(w), r) + 2B(A_u(w), A_u(w)) \ge 0.$$

Since $A_u(w) = 0$, this reduces to $B(\sigma(w), r) \ge 0$. Summing over m yields

$$B(p, r) \ge 0.$$

On the other hand, first-order criticality with $v = u$ gives

$$B(\sigma(u), r) = 0.$$

Therefore

$$B(r, r) = B(\sigma(u) - p, r) = -B(p, r) \le 0.$$

Positive-definiteness of B implies $B(r, r) \ge 0$, hence $B(r, r) = 0$, and therefore $r = 0$. □

Corollary 2.2 (Image case). *If $p \in \mathrm{im}(A_u)$, then $r(u, p) = 0$.*

Proof. Apply Lemma 2.1 with no kernel-square terms and $u_{\mathrm{img}} = u$. □

Corollary 2.3 (Constant scalar relation; Lean: `residual_eq_zero_of_constant_relation`). *If there exists $0 \ne c \in \mathbb{R}^4$ such that $\rho_u(c) = 0$, then $r(u, p) = 0$ for every SOS quartic p.*

Proof. Write $p = \sum_j q_j^2$ as in Definition 1.2. For each j, define

$$w^{(j)} = (c_0 q_j, \dots, c_3 q_j) \in \mathbb{R}[x_0, x_1]^4_{\le 2}.$$

Then $A_u(w^{(j)}) = q_j \rho_u(c) = 0$, while

$$\sigma(w^{(j)}) = \Big(\sum_{i=0}^{3} c_i^2\Big) q_j^2.$$

Since $\sum_i c_i^2 > 0$, each q_j^2 is a positive scalar multiple of a kernel square. Summing over j produces the decomposition required by Lemma 2.1. □

Proposition 2.4 (Equivariance under affine changes and orthogonal mixing; Lean families in `AffineSocpTransform.lean` and `FactorTransform.lean`). *The properties of being admissible, being an SOS quartic, being a* SOCP*, and having vanishing residual are preserved by*

1. *invertible affine changes of variables in x_0, x_1, and*
2. *orthogonal changes of basis in the four-dimensional coefficient space of the factor coordinates.*

Proof. Both operations preserve the quadratic degree bound, commute with σ, transport A_u to the corresponding linearized map in the new coordinates, and transport the bilinear form by pullback. Hence they preserve the first- and second-order criticality conditions and the equality $r(u, p) = 0$. □

3 Three global closure mechanisms

The full proof uses three conceptual closure results.

Theorem 3.1 (Span-three certificate; Lean: `residual_eq_zero_of_contains_aff1`). *Assume there exist $c^{(0)}, c^{(1)}, c^{(2)} \in \mathbb{R}^4$ such that*

$$\rho_u(c^{(0)}) = 1, \qquad \rho_u(c^{(1)}) = x_0, \qquad \rho_u(c^{(2)}) = x_1.$$

Then $r(u, p) = 0$ for every SOS quartic p.

Proof. Fix a quadratic $q \in \mathbb{R}[x_0, x_1]_{\le 2}$. Write

$$q = q_{\le 1} + q_{\text{hom}}, \qquad q_{\text{hom}} = x_0 a + x_1 b$$

with $a, b \in \mathbb{R}[x_0, x_1]_{\le 1}$. Define the raw kernel direction

$$w_{\text{raw}} = \rho(-c^{(0)}, q_{\text{hom}}) + \rho(c^{(1)}, a) + \rho(c^{(2)}, b),$$

where $\rho(c, h)$ means the admissible direction with i-th coordinate $c_i h$. By construction,

$$A_u(w_{\text{raw}}) = -q_{\text{hom}} + x_0 a + x_1 b = 0.$$

Now expand $\sigma(w_{\text{raw}})$. The quartic contribution comes entirely from the square of the first summand:

$$\sigma(w_{\text{raw}})_{\deg 4} = \Big(\sum_{i=0}^{3} (c_i^{(0)})^2\Big) q_{\text{hom}}^2.$$

Every other term contains at least one factor a or b, hence has degree at most 3. After scaling w_{raw} by

$$\alpha = \Big(\sum_i (c_i^{(0)})^2\Big)^{-1/2},$$

we obtain a kernel direction w such that

$$q^2 - \sigma(w) \in \mathbb{R}[x_0, x_1]_{\le 3}.$$

Since $1, x_0, x_1 \in W(u) := \operatorname{span}_{\mathbb{R}}\{u_0, u_1, u_2, u_3\}$, every cubic lies in $\operatorname{im}(A_u)$: multiply the exact relations for $1, x_0, x_1$ by suitable linear polynomials. Thus q^2 is an image term plus one kernel square.

Applying this construction to each quadratic summand in an SOS decomposition of p and summing gives the decomposition in Lemma 2.1. Hence $r(u, p) = 0$. ☐

Theorem 3.2 (Mixed-affine coarse theorem; Lean: `residual_eq_zero_of_relations_const_affineLine`). *Assume there exist $c^{(0)}, c^{(1)} \in \mathbb{R}^4$ such that*

$$\rho_u(c^{(0)}) = 1, \qquad \rho_u(c^{(1)}) = r + a x_0 + b x_1,$$

with $a^2 + b^2 \neq 0$. Then $r(u, p) = 0$ for every SOS quartic p.

Proof. By Proposition 2.4, we may apply an affine change of variables sending $r + a x_0 + b x_1$ to x_0. Thus it is enough to treat the normalized case where 1 and x_0 are exact relations.

The remaining two coordinates then define a quadratic plane in $\operatorname{span}_{\mathbb{R}}\{x_0^2, x_0 x_1, x_1^2\}$. There are only three possibilities.

1. The plane is surjective: an invertible basis of either $\operatorname{span}\{x_0^2, x_1^2\}$ or $\operatorname{span}\{x_0 x_1, x_0^2 - x_1^2\}$. Then every quartic lies in $\operatorname{im}(A_u)$, so Corollary 2.2 applies.

2. The plane is of rank 14, equivalently an invertible basis of $\mathrm{span}\{x_0x_1, x_1^2\}$. Then the only missing quotient class is x_0^4, and it is supplied by the explicit kernel square

$$\sigma\big(\rho(-c_2, tx_0x_1) + \rho(c_3, tx_0^2)\big)$$

for the two plane relations c_2, c_3.

3. The plane is of rank 13, equivalently an invertible basis of $\mathrm{span}\{x_0^2, x_0x_1\}$. Then the only missing quotient classes are x_1^3 and x_1^4, and both are supplied by the kernel family

$$\rho(-c_2, x_1\ell) + \rho(c_3, x_0\ell), \qquad \ell = b + cx_1.$$

In every case, each SOS square is an image term plus kernel-square corrections, so Lemma 2.1 yields $r(u,p) = 0$. Transport back by Proposition 2.4. □

Theorem 3.3 (Surjective homogeneous-basis theorem; Lean: `exact_relations_of_homQuadBasis_det` and `residual_eq_zero_of_relations_const_homQuadBasis_det`). *Assume there exists $c^{(0)} \in \mathbb{R}^4$ with $\rho_u(c^{(0)}) = 1$, and three further relations $d^{(0)}, d^{(1)}, d^{(2)} \in \mathbb{R}^4$ whose homogeneous quadratic parts form an invertible 3×3 matrix in the basis (x_0^2, x_0x_1, x_1^2). Then $r(u,p) = 0$ for every SOS quartic p.*

Proof. Invertibility of the homogeneous coefficient matrix lets us solve back for exact relations with homogeneous parts exactly x_0^2, x_0x_1, and x_1^2. Since 1 is already an exact relation, every quartic monomial lies in $\mathrm{im}(A_u)$:

- degree ≤ 2 monomials come from multiplying the relation 1 by a quadratic,
- cubic monomials come from multiplying one of x_0^2, x_0x_1, x_1^2 by a linear polynomial,
- quartic monomials come from multiplying one of x_0^2, x_0x_1, x_1^2 by a quadratic monomial.

Thus every quartic p belongs to $\mathrm{im}(A_u)$, and Corollary 2.2 applies. □

Theorem 3.4 (Low-affine plane theorem; Lean families in `RepresentativeLowAffine.lean`). *Assume there are exact relations for x_0 and x_1, together with two more quadratic relations whose homogeneous parts span a nonzero plane in $\mathrm{span}_{\mathbb{R}}\{x_0^2, x_0x_1, x_1^2\}$. Then every SOS quartic p satisfies $r(u,p) = 0$.*

Proof. The homogeneous plane is classified by the three 2×2 minors of its coefficient matrix. If the cross determinant vanishes, the plane has a common factor and reduces to the common-factor charts. If the completed-square discriminant vanishes, it reduces to the cross chart. Otherwise it reduces, after shear and scaling, to either the diagonal-sum or diagonal-difference chart. In each chart the image is explicitly described and the missing quotient classes are supplied by equally explicit kernel-square families. This is exactly the low-affine analogue of the rank-13 and rank-14 mixed-affine arguments. Applying Lemma 2.1 in each normal form proves the theorem. □

Theorem 3.5 (Affine-rank-one closure after line normalization; Lean families `residual_eq_zero_of_exactAffineDimOne_aff` `...tailRangeOne`, `...tailRangeTwo`). *Assume $\dim \mathcal{E}(u) = 1$, that $\mathcal{E}(u)$ contains a nonconstant affine line but no constant relation, and normalize that line to x_0. Let $\mathcal{T}(u)$ be the space of exact affine relations with zero x_0 coefficient, projected to their $(1, x_1)$-tails. Then $\dim \mathcal{T}(u) \in \{0, 1, 2\}$, and in each of the three cases every SOS quartic satisfies $r(u,p) = 0$.*

Proof. If $\dim \mathcal{T}(u) = 0$, all complementary relations are homogeneous, giving the linear low-affine endpoint.

If $\dim \mathcal{T}(u) = 1$, there is a common one-dimensional affine tail. After normalization one reaches a finite list of affine-rank-one charts: repeated line, shared-tail, common-factor, and diagonal endpoints. Each endpoint has an explicit quartic image theorem and an explicit kernel-square correction for the missing quotient classes.

If $\dim \mathcal{T}(u) = 2$, the tails span $\mathrm{span}\{1, x_1\}$. The corresponding tail matrix either has nonzero determinant, which is the generic range-two case, or determinant zero. In the determinant-zero regime, a shear reduction produces either the shared-tail family or the x_0^2-tail family; a discriminant then splits each of these into the zero and nonzero subcases. Every one of these finitely many endpoints is solved by an explicit image-plus-kernel-square certificate. Hence $r(u,p) = 0$ in all three tail ranks. □

4 The exact-affine classifier

We now explain the classifier that handles the linearly independent case.

Lemma 4.1 (Dimension bounds for exact affine relations)**.** *Assume* $\ker(\rho_u) = \{0\}$*. Then*

$$1 \le \dim \mathcal{E}(u) \le 3.$$

Hence $\dim \mathcal{E}(u) \in \{1, 2, 3\}$.

Proof. By Definition 1.3, $\mathcal{E}(u)$ is the kernel of the homogeneous coefficient map

$$\mathbb{R}^4 \longrightarrow \operatorname{span}_{\mathbb{R}}\{x_0^2, x_0x_1, x_1^2\} \simeq \mathbb{R}^3.$$

Since the domain has dimension 4 and the codomain has dimension 3, rank–nullity gives $\dim \mathcal{E}(u) \ge 1$. On the other hand, because $\ker(\rho_u) = \{0\}$, the affine coefficient map on $\mathcal{E}(u)$ is injective, so $\dim \mathcal{E}(u) \le \dim \mathbb{R}[x_0, x_1]_{\le 1} = 3$. □

Theorem 4.2 (Exact-affine dimension 3; Lean: `residual_eq_zero_of_exactAffineDimThree`)**.** *If* $\ker(\rho_u) = \{0\}$ *and* $\dim \mathcal{E}(u) = 3$*, then* $r(u, p) = 0$*.*

Proof. The affine coefficient map $\mathcal{E}(u) \to \mathbb{R}[x_0, x_1]_{\le 1}$ is injective and both domain and codomain have dimension 3, so it is an isomorphism. Therefore there are exact relations for $1, x_0, x_1$. Apply Theorem 3.1. □

Theorem 4.3 (Exact-affine dimension 2 with a constant relation; Lean: `residual_eq_zero_of_exactAffineDimTwo_contains`
If $\ker(\rho_u) = \{0\}$*,* $\dim \mathcal{E}(u) = 2$*, and* $1 \in \rho_u(\mathcal{E}(u))$*, then* $r(u, p) = 0$*.*

Proof. Fix an exact constant relation 1. Since $\mathcal{E}(u)$ has dimension 2, there is a second independent exact affine relation $r + ax_0 + bx_1$ with $a^2 + b^2 \ne 0$. Apply Theorem 3.2. □

Theorem 4.4 (Exact-affine dimension 2 without a constant relation; Lean: `residual_eq_zero_of_exactAffineDimTwo_noCon`
If $\ker(\rho_u) = \{0\}$*,* $\dim \mathcal{E}(u) = 2$*, and* $1 \notin \rho_u(\mathcal{E}(u))$*, then* $r(u, p) = 0$*.*

Proof. First extract an affine pair $x_0 + r_0$, $x_1 + r_1$ from $\mathcal{E}(u)$, then translate by $(x_0, x_1) \mapsto (x_0 - r_0, x_1 - r_1)$. After this translation, we have exact relations for x_0 and x_1, and every remaining exact affine relation has zero linear part.

Consider the translated zero-linear-tail kernel. There are two possibilities.

1. Every such relation has zero constant term. Then the complementary quadratic relations form a genuine homogeneous plane in $\operatorname{span}_{\mathbb{R}}\{x_0^2, x_0x_1, x_1^2\}$, so Theorem 3.4 applies.
2. Some translated kernel relation has nonzero constant term. Then choose a pair of translated relations with constants 1 and 0. If their homogeneous parts are dependent, a linear combination produces a constant relation and one falls back to Theorem 3.1. If their homogeneous parts are independent, the constant-perturbed low-affine plane is classified by the same common-factor/cross/diagonal normal forms as in Theorem 3.4, now with one relation shifted by 1. The formalized constant-perturbed low-affine plane theorems close these cases.

Thus $r(u, p) = 0$. □

Theorem 4.5 (Exact-affine dimension 1 with a constant relation; Lean: `residual_eq_zero_of_exactAffineDimOne_contains`
If $\dim \mathcal{E}(u) = 1$ *and* $1 \in \rho_u(\mathcal{E}(u))$*, then* $r(u, p) = 0$*.*

Proof. Let $c^{(0)}$ generate $\mathcal{E}(u)$ with $\rho_u(c^{(0)}) = 1$. Consider the subspace

$$K = \{c \in \mathbb{R}^4 : [1]\rho_u(c) = 0\}.$$

Since the constant coefficient map is surjective, $\dim K = 3$. The projection from K to homogeneous quadratic coefficients is injective: if a vector in K also has zero homogeneous quadratic part, then it would give a second independent exact affine relation, contradicting $\dim \mathcal{E}(u) = 1$. Thus the three vectors in K provide three independent quadratic relations.

Now inspect their affine tails. The map sending a relation in K to its (x_0, x_1)-tail has rank 0, 1, or 2. In each case, elementary row operations inside K produce a basis whose relations are of the form

$$(\text{affine tail}) + x_0^2, \qquad (\text{affine tail}) + x_0 x_1, \qquad (\text{affine tail}) + x_1^2.$$

Their homogeneous coefficient matrix is therefore invertible. Applying Theorem 3.3 yields $r(u, p) = 0$. □

Theorem 4.6 (Exact-affine dimension 1 without a constant relation; Lean: `residual_eq_zero_of_exactAffineDimOne_noCon` *If* $\ker(\rho_u) = \{0\}$*,* $\dim \mathcal{E}(u) = 1$*, and* $1 \notin \rho_u(\mathcal{E}(u))$*, then* $r(u, p) = 0$*.*

Proof. Extract the unique exact affine line and normalize it by an affine equivalence to x_0. After this normalization, the remaining exact affine relations with zero x_0-coefficient define the tail space $\mathcal{T}(u) \subseteq \operatorname{span}\{1, x_1\}$. Its dimension is 0, 1, or 2. The three cases are exactly the three tail-rank branches in Theorem 3.5, so $r(u, p) = 0$. □

Theorem 4.7 (Exact-affine classifier; Lean: `residual_eq_zero_of_relationPolyLin_ker_eq_bot`). *If* $\ker(\rho_u) = \{0\}$*, then* $r(u, p) = 0$ *for every SOS quartic* p *and every* SOCP u*.*

Proof. By Lemma 4.1, $\dim \mathcal{E}(u) \in \{1, 2, 3\}$.

- If $\dim \mathcal{E}(u) = 3$, apply Theorem 4.2.
- If $\dim \mathcal{E}(u) = 2$, split according to whether $1 \in \rho_u(\mathcal{E}(u))$ and apply Theorems 4.3 and 4.4.
- If $\dim \mathcal{E}(u) = 1$, split according to whether $1 \in \rho_u(\mathcal{E}(u))$ and apply Theorems 4.5 and 4.6.

These cases exhaust the possibilities. □

5 Proof of the main theorem

Theorem 5.1 (No spurious rank-4 SOCPs for ternary quartics; Lean: `ternaryQuartic_rankFour_no_spurious_socp`). *Let* p *be an SOS quartic, let* $u \in \mathbb{R}[x_0, x_1]^4_{\leq 2}$*, and let* B *be a positive-definite bilinear form on* $\mathbb{R}[x_0, x_1]$*. If* u *is a second-order critical point of* p*, then*

$$r(u, p) = 0.$$

Proof. Split into two cases according to $\ker(\rho_u)$.

Case 1: $\ker(\rho_u) \neq \{0\}$**.** Then there exists $0 \neq c \in \mathbb{R}^4$ with $\rho_u(c) = 0$. By Corollary 2.3, $r(u, p) = 0$.

Case 2: $\ker(\rho_u) = \{0\}$**.** Then Theorem 4.7 applies directly and gives $r(u, p) = 0$.

This proves the theorem. □

Remark 5.2. The proof has the same overall shape as the univariate argument but with a different algebraic middle layer. The univariate proof closes by a single divisibility argument on the common factor g. In the ternary quartic case, the multivariate replacement is the exact-affine classifier:

$$\text{dependent relation or } \dim \mathcal{E}(u) \in \{1, 2, 3\},$$

followed by the span-three, mixed-affine, low-affine, and affine-rank-one closure theorems described above.